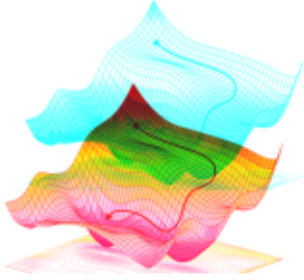
Advanced Risk Minimization

Risk Minimizers



Learning goals

- Know the concepts of the Bayes optimal model (also: risk minimizer, population minimizer)
- Bayes risk
- Consistent learners
- Bayes regret, estimation and approximation error
- Optimal constant model

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- Consistent learners

EMPIRICAL RISK MINIMIZATION

Very often, in ML, we minimize the empirical risk

$$\mathcal{R}_{\text{emp}}(f) = \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right)$$

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where

- each observation $\left(\mathbf{x}^{(i)}, y^{(i)}\right) \in \mathcal{X} \times \mathcal{Y}$, so from feature and target space
- $f_{\mathcal{H}} : \mathcal{X} \to \mathbb{R}^g$, f is a model from hypothesis space $\mathcal{H}$; maps a feature vector to output score; sometimes or often we omit $\mathcal{H}$ in the index
- each observation $\left(\mathbf{x}^{(i)}, y^{(i)}\right) \in \mathcal{X} \times \mathcal{Y}$ and $\mathcal{X}, \mathcal{Y}$ are the feature and target spaces, respectively,
- $L : (\mathcal{Y} \times \mathbb{R}^g) \to \mathbb{R}$ is loss;
- $f : \mathcal{X} \to \mathbb{R}^g$, $f \in \mathcal{H}$ is a model which maps a feature vector to a numerically encoded element of $\mathcal{Y}$ and $\mathcal{H}$ is the hypothesis space,
- $L(y, f)$ measures distance between label and prediction
- We assume that $(\mathbf{x}, y) \sim \mathbb{P}_{xy}$ and $\left(\mathbf{x}^{(i)}, y^{(i)}\right) \overset{\text{i.i.d.}}{\sim} \mathbb{P}_{xy}$
- $L : (\mathcal{Y} \times \mathbb{R}^g) \to \mathbb{R}$ is the loss function which measures the dissimilarity of the model prediction and the true target,
- $\mathbb{P}_{xy}$ is the distribution of the data generating process (DGP)

Let's define (and minimize) loss in expectation, the theoretical risk:

- and we assume that $\left(\mathbf{x}^{(i)}, y^{(i)}\right) \overset{\text{i.i.d.}}{\sim} \mathbb{P}_{xy}$ where $\mathbb{P}_{xy}$ is the distribution of the data generating process (DGP).

$$\mathcal{R}(f) := \mathbb{E}_{xy}[L(y, f(\mathbf{x}))] = \int L(y, f(\mathbf{x}))\, \mathrm{d}\mathbb{P}_{xy}$$

What is the theoretical justification for this procedure?

OPTIMAL CONSTANTS FOR A LOSS

- Let's assume some RV $Z \in \mathcal{Y}$ for a label
- Z not Y, because we want to fiddle with its distribution
- Assume Z has distribution Q, so $Z \sim Q$
- We can now consider $\arg\min_c \mathbb{E}_{Z \sim Q}[L(Z, c)]$
  so the score-constant which loss-minimally approximates Z

We will consider 3 cases for Q:

- $Q = P_Y$, simply our labels and their marginal distribution in $\mathbb{P}_{xy}$
- $Q = P_{Y|X=x}$, conditional label distribution at point $X = x$
- $Q = P_n$, the empirical product distribution for data $y_1, \ldots, y_n$

If we can solve $\arg\min_c \mathbb{E}_{Z \sim Q}[L(Z, c)]$ for any Q, we will get multiple useful results!

OPTIMAL CONSTANT MODEL

- We would like a loss optimal, constant baseline predictor
- A "featureless" ML model, which always predicts the same value
- Can use it as baseline in experiments, if we don't beat this with more complex model, that model is useless
- Will also be useful as component in algorithms and derivations

$$f_c^* = \arg\min_{c \in \mathbb{R}} \mathbb{E}_{xy}\left[L(y, c)\right] = \arg\min_{c \in \mathbb{R}} \mathbb{E}_y\left[L(y, c)\right]$$

and

$f(\mathbf{x}) = \theta = c$ that optimizes the empirical risk $\mathcal{R}_{\text{emp}}(\theta)$ is denoted as as $\hat{f}_c = \arg\min_{c \in \mathbb{R}} \mathcal{R}_{\text{emp}}(\theta)$.

OPTIMAL CONSTANT MODEL

To derive the risk-minimizer, observe that by the law of total expectation

$$R(f) = \mathbb{E}_{(\mathbf{x},y)}[L(y, f(\mathbf{x}))] = \mathbb{E}_{\mathbf{x}}\left[\mathbb{E}_{y|\mathbf{x}}[L(y, f(\mathbf{x}))] \mid \mathbf{x} = \mathbf{x}\right].$$

- Let's start with the simplest case, L2 loss
- And we want to find and optimal constant model for
- We can choose $f(\mathbf{x})$ as we want (unrestricted hypothesis space, no assumed functional form)
- Hence, for a fixed value $\mathbf{x} \in \mathcal{X}$ we can select **any** value c we want to predict. So we construct the **point-wise optimizer**

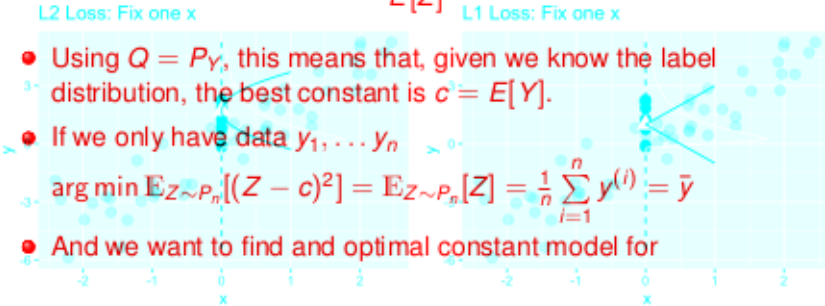
$$f^*(\mathbf{x}) = \operatorname{argmin}_c \mathbb{E}_{y|x}[L(y, c) \mid \mathbf{x} = \mathbf{x}] \quad \forall \mathbf{x} \in \mathcal{X}.$$

$$\arg\min \mathbb{E}[L(Z, c)] =$$

$$\arg\min \mathbb{E}[(Z - c)^2] =$$

$$\arg\min \mathbb{E}[Z^2] - 2cE[Z] + c^2 =$$

$$E[Z]$$



- Using $Q = P_Y$, this means that, given we know the label distribution, the best constant is $c = E[Y]$.
- If we only have data $y_1, \ldots y_n$

$$\arg\min \mathbb{E}_{Z \sim P_n}[(Z - c)^2] = \mathbb{E}_{Z \sim P_n}[Z] = \frac{1}{n}\sum_{i=1}^{n} y^{(i)} = \bar{y}$$

- And we want to find and optimal constant model for

RISK MINIMIZER AND EMPIRICAL RISK

The risk minimizer is mainly a theoretical tool:

- In practice we need to restrict the hypothesis space $\mathcal{H}$ such that
- The hypothesis space $\mathcal{H}$ is unrestricted. We can choose any $f : \mathcal{X} \to \mathbb{R}^g$.
- In practice we (usually) do not know $\mathbb{P}_{xy}$. Instead of $\mathcal{R}(f)$, we are minimizing the empirical risk
- We also assume an ideal optimizer; the risk minimization can always be solved perfectly and efficiently.
- We know $\mathbb{P}_{xy}$.

$$\hat{f} = \arg\min_{f \in \mathcal{H}} \mathcal{R}_{\text{emp}}(f) = \arg\min_{f \in \mathcal{H}} \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right)$$

Let us assume we are in an "ideal world":

How should f be chosen?

Note that according to the **law of large numbers** (LLN), the empirical risk converges to the true risk (but beware of overfitting!):

$$\bar{\mathcal{R}}_{\text{emp}}(f) = \frac{1}{n} \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right) \xrightarrow{n \to \infty} \mathcal{R}(f).$$

RISK MINIMIZER AND APPROXIMATION ERROR

The f with minimal risk across all (measurable) functions is called the **risk minimizer**, **population minimizer** or **Bayes optimal model**.

$$f^* = \arg\min_{f:\mathcal{X}\to\mathbb{R}^g} \mathcal{R}_L(f) = \arg\min_{f:\mathcal{X}\to\mathbb{R}^g} \mathbb{E}_{xy}\left[L(y, f(\mathbf{x}))\right]$$

$$= \arg\min_{f:\mathcal{X}\to\mathbb{R}^g} \int L(y, f(\mathbf{x}))\, \mathrm{d}\mathbb{P}_{xy}.$$

The resulting risk is called **Bayes risk**: $\mathcal{R}_L^* = \mathcal{R}_L(f^*)$

$$\mathcal{R}_L\left(\hat{f}\right) - \mathcal{R}_L^* = \underbrace{\left[\mathcal{R}_L\left(\hat{f}\right) - \inf_{f\in\mathcal{H}} \mathcal{R}_L(f)\right]}_{\text{estimation error}} + \underbrace{\left[\inf_{f\in\mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^*\right]}_{\text{approximation error}}$$

OPTIMAL POINT-WISE PREDICTIONS

To derive the risk minimizer, observe that by law of total expectation

$$\mathcal{R}_L(f) = \mathrm{E}_{xy}\left[L\left(y, f(\mathbf{x})\right)\right] = \mathrm{E}_x\left[\mathrm{E}_{y|x}\left[L\left(y, f(\mathbf{x})\right) \mid \mathbf{x} = \mathbf{x}\right]\right].$$

- We can choose $f(\mathbf{x})$ as we want (unrestricted hypothesis space, no assumed functional form)
- Hence, for a fixed value $\mathbf{x} \in \mathcal{X}$ we can select **any** value c we want to predict. So we construct the **point-wise optimizer**

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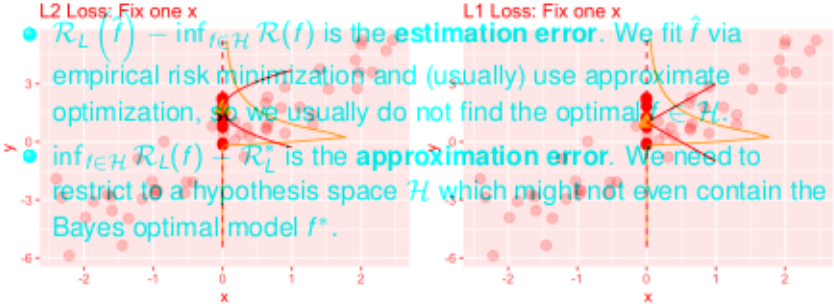

- $\mathcal{R}_L\left(\hat{f}\right) - \inf_{f \in \mathcal{H}} \mathcal{R}(f)$ is the **estimation error**. We fit $\hat{f}$ via empirical risk minimization and (usually) use approximate optimization, so we usually do not find the optimal $f \in \mathcal{H}$.
- $\inf_{f \in \mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^*$ is the **approximation error**. We need to restrict to a hypothesis space $\mathcal{H}$ which might not even contain the Bayes optimal model f^*.

THEORETICAL AND EMPIRICAL RISK

The risk minimizer is mainly a theoretical tool:

- In practice we need to restrict the hypothesis space $\mathcal{H}$ such that we can efficiently search over it.
- In practice we (usually) do not know $\mathbb{P}_{xy}$. Instead of $\mathcal{R}(f)$, we are optimizing the empirical risk

$$f_{\mathcal{H}} = \arg\min_{f \in \mathcal{H}} \mathcal{R}_{\text{emp}}(f) = \arg\min_{f \in \mathcal{H}} \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right)$$

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$$\bar{\mathcal{R}}_{\text{emp}}(f) = \frac{1}{n} \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right) \overset{n \to \infty}{\longrightarrow} \mathcal{R}(f).$$

ESTIMATION AND APPROXIMATION ERROR

Goal of learning: Train a model $\hat{f}$ for which the true risk $\mathcal{R}_L\left(\hat{f}\right)$ is close to the Bayes risk $\mathcal{R}_L^*$. In other words, we want the **Bayes regret** to be as low as possible.

$$\mathcal{R}_L\left(\hat{f}\right) - \mathcal{R}_L^*$$

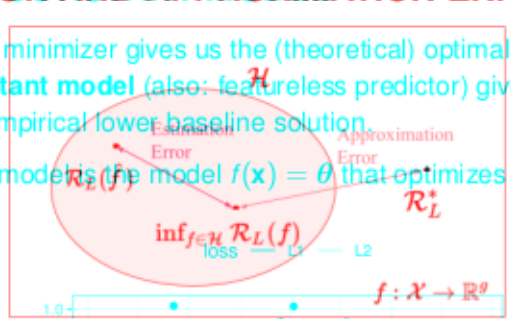
The Bayes regret can be decomposed as follows:

$$\mathcal{R}_L\left(\hat{f}\right) - \mathcal{R}_L^* = \underbrace{\left[\mathcal{R}_L\left(\hat{f}\right) - \inf_{f \in \mathcal{H}} \mathcal{R}_L(f)\right]}_{\text{estimation error}} + \underbrace{\left[\inf_{f \in \mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^*\right]}_{\text{approximation error}}$$

ESTIMATION AND APPROXIMATION ERROR / 2

While the risk minimizer gives us the (theoretical) optimal solution, the **optimal constant model** (also: featureless predictor) gives us a computable empirical lower baseline solution.

The constant model is the model $f(\mathbf{x}) = \theta$ that optimizes the empirical risk $\mathcal{R}_{\text{emp}}(\theta)$.



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